

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No 3

Aim: Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv('Daily Coffee Intake vs Sleep Duration.csv')

# Preprocess the data
# Fill missing Health_Issues values with 'None' (Assuming missing means no health issues)
df['Health_Issues'] = df['Health_Issues'].fillna('None')

# Select key features and encode categorical text (Gender, Stress_Level, Health_Issues) into numbers
features = ['Age', 'Gender', 'Coffee_Intake', 'Sleep_Hours', 'BMI', 'Heart_Rate', 'Stress_Level', 'Health_Issues']
X = pd.get_dummies(df[features], drop_first=True)
y = df['Sleep_Quality']

# Split into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Build and train the Decision Tree
# Limiting depth to 3 prevents overfitting and keeps the visual interpretable
clf = DecisionTreeClassifier(max_depth=3, random_state=42)
clf.fit(X_train, y_train)

# Evaluate the model
y_pred = clf.predict(X_test)
print(f'Accuracy: {accuracy_score(y_test, y_pred):.4f}')
print(f'Classification Report:\n', classification_report(y_test, y_pred))

# Visualize the tree
plt.figure(figsize=(20, 10))
plot_tree(clf, feature_names=X.columns.tolist(), class_names=clf.classes_.tolist(), filled=True, rounded=True, fontsize=10)
plt.title('Sleep Quality Decision Tree')
plt.show()
```

Output:

